MIL-E-1/1324A(NAVY)
12 November 1965
SUPERSEDING
MIL-E-1/1324(NAVY)
19 February 1960
(See Note 2)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE-RAY

TYPE 8MP18

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Magnetic deflection, Electrostatic focus Cathode-Ray tube, rectangular.

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb2	Eb1	Ehk	Rg1
Unit:	V	Vdc	Vdc	Vdc	Vdc	Vdc	Meg
Maximum:	6.3 ± 10%	0, -155	550	18,000	1,100	±180	1.5
Minimum:	---	---	---	7,000	-550	---	---
TEST CONDITIONS:	6.3	Adjust	300	15,000	Focus	---	---

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - Not required

E-50.2 Holding period

E-50.3.1 Preheating

FSC 5960

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METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Quality conformance inspection, part 1					
5201.2	Voltage breakdown		—	—	—	—
5201.4	Voltage breakdown (magnetic types)		—	—	—	—
5206	† Gas ratio		Gr	—	0.25	—
5101	Neck straightness	Cylinder 1.503 in. (max.) I.D.; 5 in. long	—	—	—	—
5106	Blemishes		—	—	—	—
5221	† Light output	Ib = 100 uAdc	Light	90	—	ftL
5231	Spot position		—	—	7/16	in.
5236	Zero bias anode current		—	—	—	—
5241	Grid cutoff voltage		Ec1	-33	-77	Vdc
5246	Focusing electrode voltage	Ib = 100 uAdc	—	0	450	Vdc
5251	Grid No. 1 leakage		—	—	—	—
	Quality conformance inspection, part 2					
Appendix D, 20(b)	Dimensions	See figure 1	—	—	—	—
1301	Heater current		If	540	660	mA
5101	Alinement, side terminal and base	Pin position No. 6	—	-30	+30	degrees
5216	Stray emission	Eb2 = 18,000 Vdc; Ec2 = 550 Vdc; Eb1 = 0	—	—	—	—
5223	Modulation	Ib2 = 100 uAdc	Δ Ec1	—	38	Vdc
5226	Line width A	Ib2 = 100 uAdc	width	—	.30	mm

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METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Quality conformance inspection, part 2 (Cont'd)					
5251	Heater-cathode leakage		—	—	—	—
5251	Grid No. 2 leakage		—	—	—	—
1106	Permanence of marking		—	—	—	—
1101	Base, pin, and insert secureness		—	—	—	—
	Quality conformance inspection, part 3 (Periodic check tests) (Note 1)					
5201	Focusing electrode current	Ec1 = 0	Ic2	-25	+25	uAdc
5216	Cathode illumination		—	—	—	—
1331	Direct-interelectrode capacitance		Ck to all Cg1 to all	— —	8 10	pf pf
1121	Pressure		—	—	—	—
5111	Vibration		width	—	2	mm
	Quality conformance inspection, part 3 (Life tests)					
4.7.2	Life test	Group C; Eb2 = 18,000 Vdc Ec2 = 550 Vdc; Eb1 at focus Ib2 = 25 mAdc; t = 500 hrs. (min)	—	—	—	—

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METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Quality conformance inspection, part 3 (Life tests) (Cont'd)					
4.7.3 and 4.7.3.1	Life test end points					
	Line width A	Ib2 = 75 uAdc	width	---	.30	mm
	Modulation	Ib2 = 75 uAdc	Ecl	---	38	Vdc
	Stray emission	Stray emission conditions				
	Heater-cathode leakage					
	Grid No. 1 leakage					
	Grid No. 2 leakage					

NOTES:

1. Periodic check tests shall be performed once each year or when the design of the tube is changed. Four tubes shall be tested. No failures will be allowed.

2. CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Review activities:

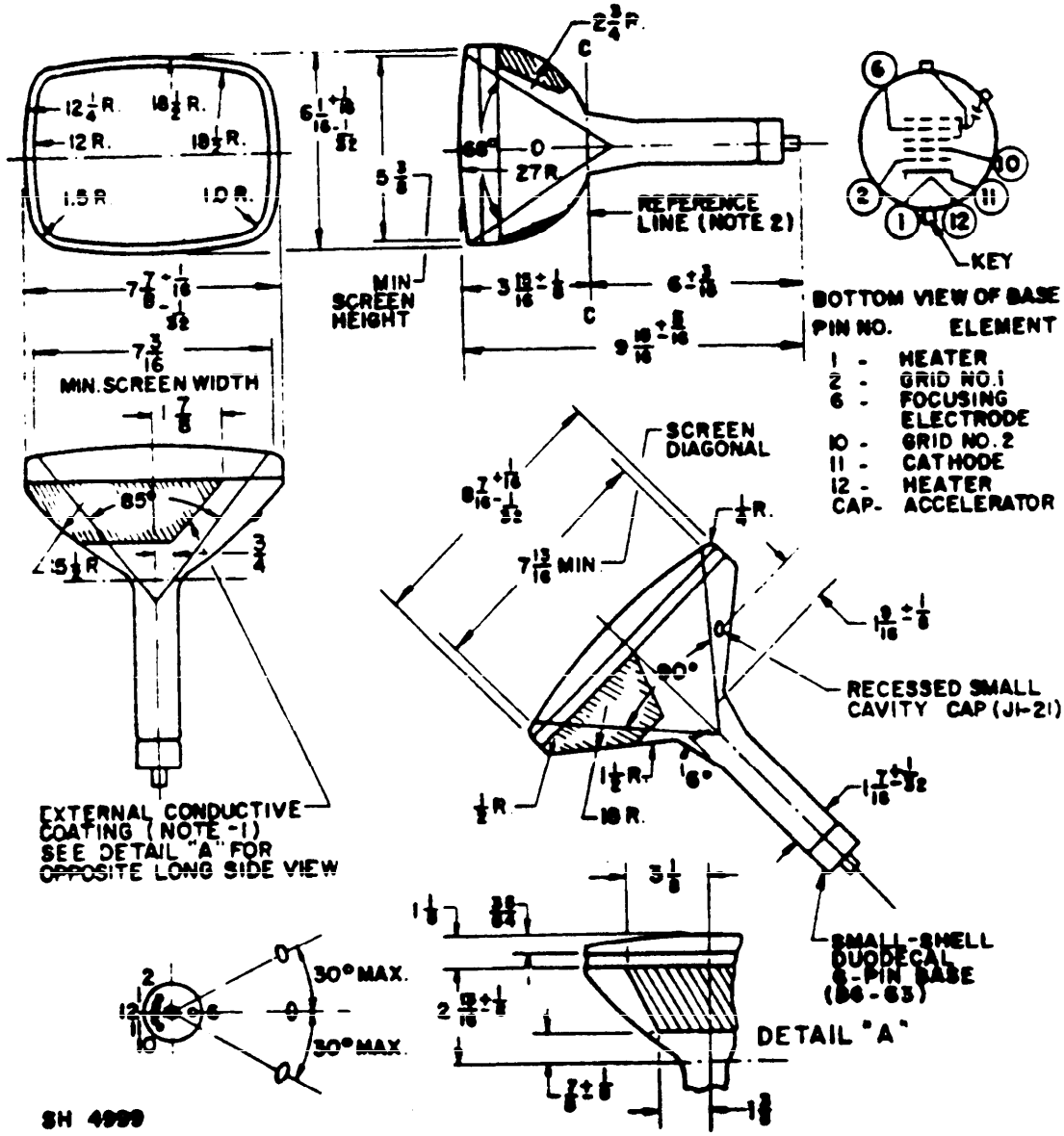
Navy - SH

User activities:

Navy - WP, MC, CG

Preparing activity:
Navy - SH
(Project 5960-N611(NAVY))

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NOTES:

1. External conductive coating must be grounded.
2. When tube neck is inserted through reference line gage JEDEC No. C-116, reference line will be determined by plane C-C when gage is resting on funnel.

Figure 1 - Outline dimensions for tube type 8MP18.

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